A.7 Log Book for Capstone Project

Mukesh Patel School of Technology Management & Engineering

Log Book for capstone project

Department: Information Technology

Program: Bachelor of Technology

Semester: VII

Project Title: RestockIQ: Demand Forecasting and Perishable Inventory Optimization for Food Retailers

Name of Faculty Mentor: Dr. Srikanth Pulipeti

Student Details

Std No.	Roll No	Name	Contact (per. email)
Student 1	A201	Yatin Patil	yatin.patil043@nmims.in
Student 2	A223	Chinmay Patil	chinmay.patil123@nmims.in
Student 3	A221	Soumy Raghuvanshi	soumy.raghuvanshi116@nmims.in

Group Members

Roll No.	Name	Sign (Students)	Grades (By Faculty Mentor)
A201	Yatin Patil		
A223	Chinmay Patil		
A221	Soumy Raghuvanshi		

Project Log Entries

Project Title: RestockIQ: Demand Forecasting and Perishable Inventory Optimization for Food Retailers

Week No.	Duration	Work Carried Out	Remarks by Faculty Mentor	Sign
1	13 Jul 2026 to 19 Jul 2026	Formed the project group and met faculty mentor Dr. Srikanth Pulipeti to finalize the capstone domain. Identified the operational challenges of daily food wastage and stockouts in artisan bakeries. Formulated the project problem statement, primary research objectives, and created the LaTeX review paper template.		

Week No.	Duration	Work Carried Out	Remarks by Faculty Mentor	Sign
2	20 Jul 2026 to 26 Jul 2026	Conducted extensive literature survey covering 25+ peer-reviewed papers on statistical time series (ARIMA, SARIMA, Prophet), machine learning (LightGBM, XGBoost), and stochastic inventory theory (Newsvendor). Identified research gaps in SME inventory systems and authored the Interim Development Plan (IDP).		
3	27 Jul 2026 to 02 Aug 2026	Acquired and ingested historical retail sales datasets, including French artisan bakery POS transactions (2021–2022) and Rossmann retail data. Conducted exploratory data analysis, audited store closure days (Mondays and Wednesdays), and established the initial Prophet baseline forecasting pipeline.		
4	03 Aug 2026 to 09 Aug 2026	Conducted requirements engineering: defined functional and non-functional requirements. Initialized Flutter cross-platform mobile repository. Scaffolded mobile views and conceptualized the operational store briefing layer powered by large language models for non-technical bakery managers.		
5	10 Aug 2026 to 16 Aug 2026	Formulated the perishable stock waste mitigation strategy: established tier-based discounting (HIGH 20%, MEDIUM 10%) and bundling rules for items nearing shelf-life expiry. Designed the decoupled system architecture separating batch forecasting jobs from interactive REST APIs.		
6	17 Aug 2026 to 23 Aug 2026	Implemented baseline forecasting models: 7-day rolling moving average, 4-week same-weekday average, and calendar-aligned seasonal naive. Evaluated baseline error benchmarks (WAPE 30.39%) across cross-validation splits to establish formal performance lower bounds.		
7	24 Aug 2026 to 30 Aug 2026	Integrated Google Gemini LLM API (gemini-3-flash-preview) for automated executive store briefings. Developed zero-math prompt engineering framework with strict temperature constraints and fallback templates to guarantee deterministic numeric grounding.		
8	31 Aug 2026 to 06 Sep 2026	Maintained repository hygiene, updated technical documentation, and audited data pipeline integrity. Prepared the academic Review-II presentation deck detailing system architecture, data models, and preliminary forecasting benchmarks.		
9	07 Sep 2026 to 13 Sep 2026	Presented academic Review-II to the departmental faculty review panel. Demonstrated core forecasting methodology, system workflow, and baseline results. Received panel approval to proceed with cloud database integration and production backend development.		

Week No.	Duration	Work Carried Out	Remarks by Faculty Mentor	Sign
10	14 Sep 2026 to 20 Sep 2026	Developed two-stage hybrid forecasting architecture: Prophet capturing macro-seasonality and holiday effects, feeding residuals to a pooled gradient-boosted decision tree (LightGBM). Evaluated SKU-level sparsity and transitioned from individual models to pooled cross-item learning.		
11	21 Sep 2026 to 27 Sep 2026	Locked the production hybrid architecture. Re-engineered Flutter UI with Material 3 design tokens, responsive cards, and dynamic state management (dashboard, item detail, waste alerts, check-in, and AI briefing). Integrated teammate pull requests.		
12	28 Sep 2026 to 04 Oct 2026	Provisioned and configured MongoDB Atlas cloud database cluster (restockiq). Designed collections for users, items, and batch forecasts. Performed repository security audit, sanitized exposed credentials, and hardened .gitignore against accidental credential leaks.		
13	05 Oct 2026 to 11 Oct 2026	Implemented full FastAPI backend service (main.py): JWT token authentication, item catalog CRUD, batch forecast ingestion, and waste alert endpoints. Built and executed end-to-end integration test suite (verify_e2e.py) validating app-to-cloud communication.		
14	12 Oct 2026 to 18 Oct 2026	Containerized FastAPI backend using Docker and deployed to Render cloud hosting. Configured GitHub Actions CI/CD workflows, automated keepalive health pings, and nightly batch forecasting scheduler. Implemented reverse-proxy rate limiting.		
15	19 Oct 2026 to 26 Oct 2026	Executed rigorous empirical validation: 38-fold weekly backtesting (Model A 26.06% pooled WAPE, Canonical Hybrid 22.51% summer folds), 200-iteration circular weather placebo test (weather rejected), and 68,883-unit Newsvendor waste simulation. Passed 55/55 test suite across all 5 tiers. Resolved DEF-001 through DEF-007. Compiled Black Book and finalized submission freeze v1.0-submission.		

Date: _____ Place: Shirpur	_____ **Dr. Srikanth Pulipeti** Faculty Mentor Department of Information Technology, MPSTME, Shirpur